CISCO PACKET TRACER
PROJECT DOCUMENTATION

Implement VLANs and Trunking Using IPv6

A two-switch Layer 2 network configuration

Platform	Cisco Packet Tracer
Devices	Two Cisco switches and four end devices
Core topics	IPv6 addressing, VLANs, 802.1Q trunking, native VLAN, management SVI

1. Project Overview

This project demonstrates VLAN segmentation and trunk configuration using IPv6 addressing in Cisco Packet Tracer. Two switches, S1 and S2, are connected through their GigabitEthernet 0/1 interfaces. Staff and Management devices are placed in separate VLANs so that each department forms its own Layer 2 broadcast domain.

A dedicated Parking_Lot VLAN is used for unused switch ports, while VLAN 888 is configured as the native VLAN on the inter-switch trunk. Management IPv6 addresses are configured on VLAN 25 switched virtual interfaces (SVIs).

2. Objectives

• Create and name VLANs on both switches.

• Assign access ports to the Staff and Management VLANs.

• Configure an 802.1Q trunk between S1 and S2.

• Set VLAN 888 as the native VLAN and restrict the trunk's allowed VLAN list.

• Configure IPv6 addresses on end devices and management SVIs.

• Place unused interfaces in VLAN 88 and administratively shut them down.

• Verify VLAN, trunk, and IPv6 interface status.

3. Network Devices and Topology

Device	Quantity	Role
Cisco switches (S1 and S2)	2	Interconnect VLANs and carry traffic over a trunk
Staff PCs	2	End devices in VLAN 15
Management PCs	2	End devices in VLAN 25

Inter-switch connection: S1 GigabitEthernet0/1 connects to S2 GigabitEthernet0/1. End-device connections should use access ports in the VLAN assigned to the relevant department.

The exact PC-to-switch port numbers were not specified in the supplied notes. Use Fa0/1–Fa0/6 for Staff devices and Fa0/7–Fa0/12 for Management devices, as appropriate to the physical topology.

4. VLAN Plan

VLAN ID	VLAN Name	Purpose
15	Staff	Staff department devices
25	Management	Management devices and switch management SVIs
88	Parking_Lot	Unused switch ports; ports are shut down
888	Native	Native VLAN for the inter-switch trunk

5. IPv6 Addressing Plan

Device / Interface	VLAN	IPv6 address / prefix
Staff-PC1	15	2001:DB8:ACAD:15::A/64
Staff-PC2	15	2001:DB8:ACAD:15::B/64
Management-PC1	25	2001:DB8:ACAD:25::A/64

Device / Interface	VLAN	IPv6 address / prefix
Management-PC2	25	2001:DB8:ACAD:25::B/64
S1 – VLAN 25 SVI	25	2001:DB8:ACAD:25::1/64
S2 – VLAN 25 SVI	25	2001:DB8:ACAD:25::2/64

The IPv6 addresses use documentation prefix 2001:DB8::/32. No default gateway was included in the supplied configuration. A default gateway is not needed for communication within the same IPv6 subnet, but routing is required for communication between different VLAN subnets.

6. Switch Initial Configuration

Apply the dual-stack SDM template on both switches if the switch model and Packet Tracer IOS image support it. The switch must be reloaded for the SDM template change to take effect.

Switch S1

```
enable
configure terminal
hostname S1
sdm prefer dual-ipv4-and-ipv6 default
end
copy running-config startup-config
reload
```

Switch S2

```
enable
configure terminal
hostname S2
sdm prefer dual-ipv4-and-ipv6 default
end
copy running-config startup-config
reload
```

After the reload, continue with the VLAN, interface, trunk, and IPv6 configuration.

7. VLAN Creation on Both Switches

Create the same VLANs on S1 and S2 so that the VLANs exist at both ends of the trunk.

```
enable
configure terminal

vlan 15
 name Staff
vlan 25
 name Management
vlan 88
 name Parking_Lot
vlan 888
 name Native

end
show vlan brief
```

8. Access Port Configuration

Configure the same access-port ranges on each switch. The supplied notes assign Fa0/1–Fa0/6 to Staff, Fa0/7–Fa0/12 to Management, and Fa0/13–Fa0/24 plus G0/2 to the Parking_Lot VLAN.

Switch S1 and Switch S2

```
configure terminal

interface range fa0/1-6
```

```
 switchport mode access
 switchport access vlan 15

interface range fa0/7-12
 switchport mode access
 switchport access vlan 25

interface range fa0/13-24, g0/2
 switchport mode access
 switchport access vlan 88
 description Unused Ports
 shutdown

end
show vlan brief
```

Confirm that G0/1 is not included in the unused-port range because it is the inter-switch trunk interface.

9. Trunk Configuration

Configure GigabitEthernet0/1 as a trunk on both switches. VLANs 15, 25, and 888 are allowed across the trunk, and VLAN 888 is set as the native VLAN.

Switch S1

```
configure terminal
interface gigabitEthernet0/1
 description Connected to S2 G0/1
 switchport mode trunk
 switchport trunk native vlan 888
 switchport trunk allowed vlan 15,25,888
end
copy running-config startup-config
show interfaces trunk
```

Switch S2

```
configure terminal
interface gigabitEthernet0/1
 description Connected to S1 G0/1
 switchport mode trunk
 switchport trunk native vlan 888
 switchport trunk allowed vlan 15,25,888
end
copy running-config startup-config
show interfaces trunk
```

VLAN 88 is intentionally not included in the allowed trunk list because it is used locally for unused, shut-down ports. The native VLAN should match on both ends of the trunk.

10. Management SVI IPv6 Configuration

Assign an IPv6 address to VLAN 25 on each switch. The SVI provides an IPv6 management interface for the switch.

Switch S1

```
configure terminal
interface vlan 25
 ipv6 address 2001:DB8:ACAD:25::1/64
 no shutdown
end
copy running-config startup-config
```

Switch S2

```
configure terminal
interface vlan 25
 ipv6 address 2001:DB8:ACAD:25::2/64
 no shutdown
end
copy running-config startup-config
```

For an SVI to reach the up/up state, VLAN 25 must exist and have at least one active Layer 2 port in that VLAN, or an active trunk carrying VLAN 25. The trunk and its far-end connection must also be operational.

11. End-Device IPv6 Configuration

Configure the following IPv6 addresses manually in each PC's Desktop > IP Configuration > IPv6 Configuration settings. Use a /64 prefix length.

PC	IPv6 address	Prefix length
Staff-PC1	2001:DB8:ACAD:15::A	64
Staff-PC2	2001:DB8:ACAD:15::B	64
Management-PC1	2001:DB8:ACAD:25::A	64
Management-PC2	2001:DB8:ACAD:25::B	64

12. Verification and Testing

Run the following commands on both switches to verify the configuration:

```
show vlan brief
show interfaces trunk
show ipv6 interface brief
show running-config
show interfaces status
```

Suggested connectivity tests:

```
From Staff-PC1:
ping 2001:DB8:ACAD:15::B

From Management-PC1:
ping 2001:DB8:ACAD:25::B
ping 2001:DB8:ACAD:25::1
ping 2001:DB8:ACAD:25::2
```

Expected results: Staff-PC1 should be able to reach Staff-PC2 across the trunk. Management-PC1 should be able to reach Management-PC2 and, when the SVI is operational, the management IPv6 addresses of S1 and S2. Successful communication between VLAN 15 and VLAN 25 requires a Layer 3 routing device and appropriate IPv6 routing configuration.

13. Important Notes and Troubleshooting

• **SDM template:** Some switch models or Packet Tracer versions may not support the exact dual-stack SDM command. Check the available options with `sdm prefer ?` before applying it.

• **IPv6 routing:** The command `ipv6 unicast-routing` enables IPv6 routing on supported Layer 3 devices. It is not required simply to assign IPv6 addresses to Layer 2 switch SVIs. These two-switch configurations do not provide inter-VLAN routing.

• **Trunk mismatch:** Ensure both ends use trunk mode, native VLAN 888, and the same allowed VLAN list.

• **SVI down:** Confirm VLAN 25 exists, the trunk is up, VLAN 25 is allowed on the trunk, and at least one VLAN 25 port is active.

• **IPv6 format:** Use the double colon in the supplied host addresses, for example 2001:DB8:ACAD:15::B.

• **Command spelling:** Use `switchport mode access` and `sdm prefer`; the supplied notes contained minor typing errors.

14. Learning Outcomes

• Understanding IPv6 address structure and /64 prefix configuration.

• Creating VLANs and assigning access interfaces.

• Configuring and verifying 802.1Q trunk links.

• Understanding native VLAN operation.

• Configuring management SVIs with IPv6 addresses.

• Securing unused switch ports by placing them in a parking-lot VLAN and shutting them down.

• Using Cisco IOS show commands to verify switch configuration.

15. Conclusion

This project provided practical experience configuring VLAN segmentation and trunking in a dual-switch Cisco Packet Tracer topology using IPv6 addressing. It demonstrates core Layer 2 switching skills, management SVI configuration, native VLAN configuration, and basic unused-port security. The completed topology can be used as a learning and portfolio project for entry-level networking and Network Support roles.